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LOCALIZING PERSONA REPRESENTATIONS IN LLMS

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ABSTRACT

We present a study on how and where personas – defined by distinct sets of human characteristics, values, and beliefs – are encoded in the representation space of large language models (LLMs). Using a range of dimension reduction and pattern recognition methods, we first identify the model layers that show the greatest divergence in encoding these representations. We then analyze the activations within a selected layer to examine how specific personas are encoded relative to others, including their shared and distinct embedding spaces. We find that, across multiple pre-trained decoder-only LLMs, the analyzed personas show large differences in representation space only within the final third of the decoder layers. We observe overlapping activations for specific ethical perspectives – such as moral nihilism and utilitarianism – suggesting a degree of polysemy. In contrast, political ideologies like conservatism and liberalism appear to be represented in more distinct regions. These findings help to improve our understanding of how LLMs internally represent information and can inform future efforts in refining the modulation of specific human traits in LLM outputs. ***Warning**: This paper includes potentially offensive sample statements.*

1 Introduction

Understanding the mechanisms by which large language models (LLMs) process information, store knowledge, and generate outputs remain key open questions in research [1, 2]. One crucial and largely underexplored aspect of these models is how they encode human personality traits, ethical views, or political beliefs – often broadly referred to as *personas* [3]. Clearly understanding the mechanics of personas is important for a variety of reasons. Personas are often used to define the personality or perspective the LLM model should adopt when interacting with users [4], e.g., by prompting "Suppose you are a person who . . . " followed by a description of a particular trait or belief. For instance, if the prompt states ". . . is highly agreeable", the model may generate more cooperative and empathetic responses. If the prompt states ". . . subscribes to the moral philosophy of utilitarianism", the model's outputs may prioritize maximizing overall well-being when making ethical decisions. This can significantly influence language generation by setting a tone appropriate for the context (e.g., empathetic or professional) and by affecting behavior and reasoning capabilities [5]. Personas have also been leveraged in tasks such as synthetic data generation [6] and decision-making solutions such as LLM-as-a-judge [7]. Personas have been debated in multiple social, political, and behavioral studies as a potential replacement for human participants [8, 9, 10]. While there are interesting applications, the use of LLMs in studies raises key concerns that require careful attention [11]. Personas can enhance user experience and engagement by making models more relatable and context-aware [12, 13, 14], however, they have also been used to bypass safety measures or to trigger unintended consequences rooted in underlying biases [15], raising significant ethical and security concerns [16].

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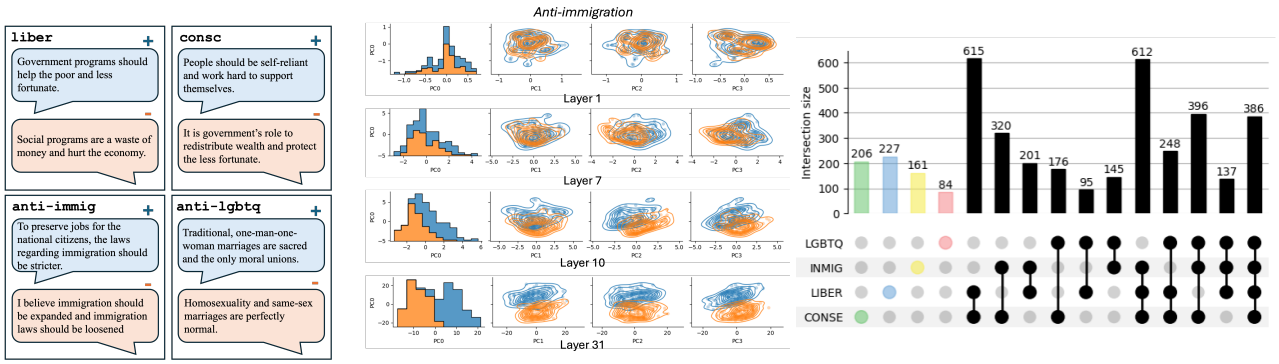

(a) Persona-specific datasets. (b) PCA of activations. (c) Overlap of salient sets of activations.

Figure 1: Overview of our study for *Llama3* on persona topic *Politics*. **(a)** Examples of MATCHINGBEHAVIOR (+) and NOTMATCHINGBEHAVIOR (–) persona statements. **(b)** PCA of representations for + and - sentences for *anti-immigration* **(Q1)**. **(c)** Overlap of sets of salient last-layer activations from + sentences, as identified by Deep Scan, across personas **(Q2)**.

Unlike traditional fair-ML decision-making frameworks, which optimize explicitly defined objectives (e.g., maximizing utility subject to a fairness metric [17, 18]), LLMs learn their decision patterns from massive, largely uncurated text corpora [19, 20], rendering their latent moral or political predispositions opaque. It is possible to refine an LLM's behavior via supervised fine-tuning on small, carefully curated datasets [21, 22]. However, the inherently open-ended nature of linguistic output makes it difficult to anticipate and constrain every downstream use case. If an LLM contains a latent political bias or moral preference that goes undetected, it may systematically privilege certain viewpoints or value systems over others, potentially amplifying polarizing or discriminatory content in high-stakes settings [9, 23, 24, 25]. Research has indeed identified gender biases in LLMs, which favor males' perspectives [26], and political biases favoring liberal viewpoints [9]. These biases are also extended to personality, morality, and stereotypes [23, 27].

Responsible AI governance demands both transparency and explainability ("What does the model encode?") as well as controllability ("How can we steer or constrain its outputs?") [28, 25].Consequently, there has been a growing call to better understand how and where personas are encoded within an LLM's internal representations [29, 30, 31, 32]. Such insights can improve the model's interpretability, transparency, and inform methods to align LLM outputs with human values [32, 33].

In this work, we aim to better understand where personas are encoded in the internal representations of LLMs (see Fig. 1 for an overview). We rely on a publicly available collection of model-generated personas [3] specifically focusing on three categories, spanning human identity and behavior: *Politics*, which include ideological leanings and political affiliations, reflecting individuals' values and societal preferences (e.g., liberal, conservative); *Ethics*, which captures moral reasoning and value-based judgments, central to human decision-making and social interactions (e.g., deontology, utilitarianism); and *Primary Personality Traits* (*Personality*), based on the Big Five model [34, 35], which provides a comprehensive framework for understanding human behavior and interpersonal dynamics (e.g., agreeableness, conscientiousness). These personas span a wide range of values, beliefs, and social preferences, providing a grounded basis for studying how LLMs encode complex human attributes [35].

We feed statements associated with different personas (see Fig. 1a) into various LLMs and extract their internal representations (i.e., activation vectors). We then analyze these representations to address the following two questions:

- **(Q1)** Where in the model are persona representations encoded? Specifically, which layers in the LLM exhibit the strongest signals for encoding persona-specific information (see Fig. 1b)?
- **(Q2)** How do these representations vary across different personas? In particular, are there consistent, uniquely activated locations within a given LLM layer where distinct persona representations are encoded (see Fig. 1c)?

This approach enables us to systematically investigate how LLMs process and differentiate persona-related information.

Our main findings are:

- The final third segment of layers (across *Llama3-8B-Instruct* (*Llama3*), *Granite-7B-Instruct*, and *Mistral-7B-Instruct*) captures the most variance in persona representations, with the last layers exhibiting the strongest separability along principal components. This suggests that higher-level semantic abstractions related to human values are encoded in later layers of the model families. *Llama3* exhibits the largest separation across layers and personas, with the last layer showing the clearest separation.

- For embeddings in *Llama3*'s last layer, we find that personas of different ethical values exhibit the highest activation overlap (17.6% of the embedding vector), while personas with different political beliefs have the most uniquely associated activations (2.1%–5.5%). This suggests that political views are more distinctly localized, while ethical views are more polysemous, sharing activations across multiple concepts.

The remainder of the paper is structured as follows: § 2 reviews related work; § 3 details the study design, dataset, and models; § 4 describes the methods used for our analysis; § 5 presents our empirical results and associated discussion; and § 6 summarizes the paper, identifies limitations, and provides directions for future work.

2 Related Work

Personas. LLM personas have attracted increasing attention for their ability to personalize and steer outputs, particularly in the development of trustworthy models [36, 37]. A persona is a natural language portrayal of an imagined individual belonging to some demographic group or reflecting certain personality traits [38, 39]. Research has found that personas can induce model outputs that are toxic and propagate stereotypes [23, 40, 41], and contain more extreme views with respect to Western vs. non-Western norms [42]. Moreover, personas have been (mis)used to circumvent built-in safety mechanisms by instructing models to adopt specific roles, e.g., that of fictional characters [43]. Understanding how LLMs encode personas is essential for harm mitigation methods [44], aligning models with diverse beliefs [10], and tailoring outputs to users' preferences.

Behavior and Value Encoding in LLMs. Early work argues that representations in BERT [45] can reveal the implicit moral and ethical values embedded in text [46]. These studies focus on quantifying deontological ethics, which determine whether an action is intrinsically right or wrong by analyzing output embeddings. We extend these ideas to a broader spectrum of human values, beliefs, and traits consisting of 14 different personas. Furthermore, we look at internal representations rather than output embeddings.

More recently, understanding an LLM's representations at different layers and token embeddings has gained increased attention [29, 47, 48], aiming to understand how concepts are represented within an LLM's (decoder) neural network, e.g., by generating human-understandable translations of information encoded in hidden representations [47]. In the context of human behavior, prior work has shown how neural activity across all layers can build feature vectors for honest and dishonest behavior detection [29]. In contrast, our work focuses on identifying subsets of the activation vector that are most representative of a given persona.

The work most related to ours [32], performed (parallel to us) a layer-wise analysis of how LLMs encode three Big Five traits by training supervised classifiers on last-token embeddings and using layer-wise perturbations to edit expressed personalities. In contrast, we (i) probe a broader set of moral, personality, and political personas, (ii) identify the minimal subset of embedding dimensions in each layer that drives each persona, and (iii) quantify overlaps between these persona embedding subsets. We link those overlaps to a phenomenon that is often described as polysemanticity, where individual neurons respond to mixtures of seemingly unrelated inputs, which affects interpretability and impacts the generation process [49, 50]. We further show that a classifier using only the activations corresponding to our identified subset of embedding dimensions yields the same test performance as one using the full embedding, thus validating the importance of this subset in encoding a persona. While we do not edit embeddings in this work, these findings open the door for more efficient, fine-grained interventions.

Deep Scan. Deep Scan has been predominantly used to detect anomalous samples in various computer vision, text, and audio tasks by analyzing patterns in neural networks [51, 52, 53]. Recent work has also taken initial steps in exploring which subsets of activations are most responsible for encoding harmful concepts, such as toxicity [51]. We build on this work and extend it by focusing exclusively on measuring the localization consistency and uniqueness of activations that encode human beliefs, values, and traits as personas in LLMs. Our study offers a systematic framework for localizing persona representations and their interactions in LLMs.

3 Study Design

This section outlines the study design, including the socio-technical motivation (§ 3.1), the dataset selection and assumptions (§ 3.2), the models used (§ 3.3), and the motivation for our research questions (§ 3.4).